R Studio

Statistical Analysis

A histogram created in R (Version 4.6.0) within the RStudio integrated development environment was used to analyze the distribution of sample sizes among the 11 papers that were part of the systematic review. Participant numbers were taken straight from the final filtered dataset, and sample size was handled as a continuous quantitative variable. The analysis reflected the initial sample sizes reported by each study because no transformations or imputations were carried out. The base R hist() function was used to create the histogram, and sample sizes were divided into equal-width intervals of ten participants (0–10, 10–20, 20–30, 30–40 and 40–50 . This interval width was chosen because it preserved visual clarity within a comparatively small sample while offering enough detail to spot distributional trends.

The histogram summarizes the frequency distribution of participant numbers across the included studies and functions as a type of descriptive statistical analysis. The histogram shows the general form of the distribution, including clustering, dispersion, skewness, and the presence of possible outliers, in contrast to measurements like the mean or median, which simply offer one summary statistic. This makes it possible to evaluate participant recruitment across the literature in a more thorough manner. The resulting distribution is positively skewed (right-skewed), with the majority of studies concentrated in the smaller sample size intervals and fewer studies represented as sample size increases. Only two studies had more than thirty participants, whilst the majority had fewer than thirty. There were no studies with sample sizes between 40 and 47 individuals, and the biggest sample size in the final dataset was found in one study with 48 participants. Overall, the distribution shows that the majority of the current body of information still comes from small-scale research.

Figure Significance

By showing how participant recruitment is dispersed throughout the body of existing literature, Figure 12 offers important insight into the methodological features of the studies included in this systematic review. The histogram shows that most studies enrolled fewer than 30 people, and the vast majority recruited relatively small participant cohorts. Only two studies recruited more than 30 participants, and one study had more than 70 participants. Several studies had less than ten participants. Many of the methodological and practical difficulties that arise when researching brain-computer interface (BCI) technology are reflected in this distribution. Participants with diseases like spinal cord damage, stroke, phantom limb pain, Parkinson's disease, or limb amputation are often recruited for studies examining neurological rehabilitation. Recruitment is time-consuming and resource-intensive since these patient populations are frequently small, geographically distributed, and subject to stringent eligibility requirements. The number of participants that can be reasonably enrolled in a single study is further limited by the fact that many BCI interventions call for specialized laboratory equipment, repeated EEG recording sessions, neurofeedback training, robotic rehabilitation devices, or functional electrical stimulation. The nascent stage of BCI research is also reflected in the prevalence of small sample sizes. Proof-of-concept studies, pilot trials, feasibility studies, and early-phase clinical investigations make up a large portion of the literature. These studies are generally intended to assess safety, feasibility, and preliminary efficacy rather than demonstrate definite clinical effectiveness. Before moving on to larger randomized controlled trials, these investigations purposefully enlist smaller participant groups. Consequently, the observed distribution indicates that many therapies are still in very early phases of clinical validation, despite the field's optimistic findings.

The distribution has significant methodological ramifications for how the entire body of information is interpreted. Smaller sample sizes lower statistical power and raise the possibility that random variation or participant characteristics could have an impact on observed outcomes. Additionally, they make it more difficult to conduct subgroup analysis or extrapolate results to larger patient populations. On the other hand, the fact that the dataset contains a number of larger trials suggests that researchers are progressively carrying out more thorough studies that can yield greater proof of the efficacy of BCI-assisted rehabilitation. The unequal distribution of participant recruitment throughout the literature is further highlighted by the lack of studies within specific sample size ranges. The histogram indicates that researchers usually undertake either relatively small exploratory investigations or significantly bigger clinical trials, with relatively few studies occupying the intermediate region, rather than showing a progressive increase in study size. This pattern can be a result of variations in study goals, institutional resources, and funding availability. Overall, Figure 12 shows that although BCI research has grown significantly, larger, multicenter clinical trials with more diverse participant populations should be the focus of future studies. Increasing sample numbers will strengthen the evidence for the clinical efficacy and long-term application of BCI technologies across various neurological groups, as well as improve statistical precision and external validity.

Connection to Included Articles

The distribution shown in Figure 12 closely reflects the methodological characteristics of the studies included in this systematic review. A number of studies were purposefully created as exploratory clinical trials, pilot studies, or feasibility studies, which usually enlist very small participant cohorts. For instance, Training with Brain-Machine Interfaces, Visuotactile Feedback, and Assisted Locomotion Improves Sensorimotor, Visceral, and Psychological Signs in Chronic Paraplegic Patients and Long-Term Training with a Brain-Machine Interface-Based Gait Protocol Induces Partial Neurological Recovery in Paraplegic Patients only recruited seven and eight participants, respectively, indicating the difficulties of conducting intensive BCI rehabilitation studies in specialized patient populations. The review's larger studies show how BCI therapies are gradually moving toward more thorough clinical evaluation. While DiSCIoser, New Approaches Based on Non-Invasive Brain Stimulation and Mental Representation Techniques Targeting Pain in Parkinson's Disease Patients, Brain–Computer Interface-Based Robotic End Effector System for Wrist and Hand Rehabilitation, and Effects of Brain–Computer Interface-Controlled Functional Electrical Stimulation Training on Shoulder Subluxation for Patients with Stroke recruited between 20 and 32 participants. These studies demonstrate the growing efforts to assess BCI-assisted rehabilitation with larger and more reliable clinical samples. Overall, Figure 12 emphasizes a key feature of the existing literature on BCI: the difficulty of recruiting participants is still hampered by the complexity of interventions, the need for specialist equipment, and the scarcity of neurological populations that qualify. However, the existence of both smaller exploratory research and a number of bigger randomized controlled trials indicates that the discipline is gradually moving toward more thorough clinical examination. The evidence for the efficacy and therapeutic translation of brain-computer interface technology will be further strengthened by future studies with larger and more varied participant populations.

Methodology and Code

In addition to providing a visual summary of participant recruitment within the present body of research, Figure 12 aimed to investigate the distribution of sample sizes across all studies included in the final systematic review. Since sample size affects statistical power, the accuracy of predicted effects, and the generalizability of study findings, understanding the distribution of sample sizes is crucial to the synthesis of evidence. The histogram adds methodological context to the qualitative synthesis of the literature by displaying the frequency of participant numbers across the included studies. The final filtered dataset created following the systematic review procedure provided the data utilized to create the histogram. Eleven studies that satisfied the predetermined inclusion criteria were kept for analysis after title, abstract, and full-text screening. The provided sample size (total number of participants) for each trial was taken straight out of the screening spreadsheet. A dataset of eleven observations was produced, with each publication contributing a single numerical value that represented the total number of participants. Prior to analysis, the retrieved values were checked for accuracy and consistency against the original screening documents. The graphic authentically depicts the participant numbers given by the original research because no observations were eliminated, altered, normalized, or calculated. R (Version 4.6.0) was used in the RStudio integrated development environment for statistical analysis and figure creation. R was chosen because it offers a repeatable computing environment where each analytical step may be recorded using code instead of being done by hand. This method increases transparency, reduces transcribing errors, and enables other researchers to precisely replicate the image using the same dataset and script. In order to verify that every observation was properly imported and matched the values found in the filtered dataset, the sample size values were input into R as a numerical vector.

To analyze the frequency distribution of sample sizes, a histogram was created using the base R hist() function. A histogram is specifically made to summarize the distribution of continuous numerical values, in contrast to categorical visualizations like bar charts. Important descriptive features of the data, such as the concentration of observations, variability, skewness, distributional gaps, and the existence of possible outliers, can be assessed by researchers. Because of this, a histogram was thought to be the best visual aid for analyzing participant recruitment in all of the included research. The ranges 0–10, 10–20, 20–30, 30–40 and 40–50 were created by building the histogram using equal-width intervals (bins) of ten participants. When creating the picture, a number of interval widths were taken into account; nevertheless, a 10-participant interval offered the best comprehensible depiction of the data. Narrower intervals would have produced many sparsely populated bins, making the distribution more fragmented and more challenging to understand because the systematic review only included eleven research. Wider intervals, on the other hand, would have made it harder for the figure to discern significant differences between research with various sample sizes. Thus, the chosen interval width struck a balance between adequate analytical information and readability. The finished histogram was produced in accordance with the visual guidelines set for each of the project's figures. The project's blue colour scheme (#1F77B4) was used to display bars with white borders to enhance the visual differentiation between neighbouring intervals. To guarantee that the visualization could be understood without reference to the accompanying text, marked axes, a descriptive figure title, and consistent font were included. The finished figure was exported as a high-resolution PNG file that could be used in the presentation materials and final report. The original sample size dataset, the R script used to create the histogram, the generated graphic, and the approach documentation were all kept in order to guarantee reproducibility. Keeping these resources in a version-controlled GitHub repository promotes transparency and adherence to best practices in reproducible research by allowing other researchers to do the analysis using the same dataset and computational approach.

Code

The histogram was generated in **R** using the following script:

```
sample_sizes <- c(
  20, 8, 8, 30, 1,
  32, 14, 8, 7, 48, 21
)
```

```
par(family = "Helvetica")

hist(
  sample_sizes,
  breaks = seq(0, 50, by = 10),
  col = "#1F77B4",
  border = "white",
  main = "Distribution of Participant Sample Sizes Across Included Studies",
  xlab = "Participant Sample Size",
  ylab = "Number of Studies",
  xaxt = "n"
)

axis(
  side = 1,
  at = seq(0, 50, by = 10),
  labels = seq(0, 50, by = 10)
)

abline(h = 0:5, col = "lightgray", lwd = 1)
```

Excel
Methodology and Code

The histogram was made using Microsoft Excel to examine how participant sample sizes were distributed throughout the 11 research that made up this systematic review. Following the application of the inclusion and exclusion criteria, participant sample sizes were taken from the final screened dataset and placed into a single spreadsheet column. Eleven observations were included in the dataset, with each study contributing one participant sample size value. Before being visualized, the extracted numbers were checked for accuracy and consistency against the source publications. Since a histogram is a suitable descriptive statistical technique for showing the distribution of continuous numerical data, it was chosen. A histogram, in contrast to summary statistics like the mean or median, shows the frequency of observations within predetermined ranges, making it easier to spot patterns in the data, such as clustering, variability, skewness, and possible distribution gaps. In addition to offering further methodological background for analyzing the data, this gives a visual summary of participant recruitment across the included research. Excel's built-in histogram chart was used to create the histogram. Ten subjects were divided into equal-width intervals (bins) of 0–10, 10–20, 20–30, 30–40, and 40–50. This interval width was chosen because it maintained a clear and understandable display for the comparatively small dataset while offering enough detail to detect significant variations in participant recruitment. The number of studies whose participant sample sizes fall within a specified range is shown by each bar. To enhance readability and preserve uniformity across all figures, the figure was designed in accordance with the standard project specifications utilizing a blue colour scheme, named axes, horizontal gridlines, and a descriptive title. The finished histogram shows the methodological features of the present body of data and offers a quantitative assessment of participant recruitment across the included studies. To promote reproducibility and transparency, the approach documentation, Excel spreadsheet, output figure, and underlying dataset were kept. The histogram was created using Microsoft Excel's built-in statistical chart features. The development of the figure didn't involve any special programming code. The Excel file used to create the histogram, the exported figure, the associated methodological description, and the participant sample size dataset are all included in the GitHub repository.